

Torsion Phenomena in Chromatic Hypergraph Cohomology

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The Chromatic Cochain Complex

Given a hypergraph $\mathcal{H}$ a state $[\mathcal{H} : s]$ is a spanning subgraph with hyperedges $s \subseteq \mathcal{H}$. The set of connected components of a state $[\mathcal{H} : s]$ is denoted by $K([\mathcal{H} : s])$. An enhanced state (s, ℓ) is a labeling $\ell : K([\mathcal{H} : s]) \rightarrow \mathcal{A}_m$ on the connected components of $[\mathcal{H} : s]$ with homogeneous elements from the algebra $\mathcal{A}_m = \mathbb{Z}[x]/(x^m)$.

The chromatic chochain group $C^{i,j}(\mathcal{H})$ is the free abelian group generated by enhanced states (s, ℓ) with conditions:

$$C^{i,j}(\mathcal{H}) := \mathbb{Z} \left\langle (s, \ell) : |s| = i, \sum_{e \in s} \deg(\ell(e)) = j \right\rangle.$$

i is the homological grading and j is the quantum grading.

Given a linear order of the hyperedges, the coboundary is

$$\delta^{i,j}(s, \ell) := \sum_{f \in \mathcal{H} \setminus s} (-1)^{b(f,s)} (s \cup \{f\}, \ell_f),$$

where $b(f, s)$ is the number of hyperedges in s which come before f in the ordering, and $\mathcal{H} \setminus s$ is exactly the set of hyperedges of $\mathcal{H}$ not in s . The labeling ℓ_f is given by the following rule

$$\ell_f(\Gamma) := \begin{cases} \ell(\Gamma \setminus \{f\}), & \Gamma \setminus \{f\} \in K[\mathcal{H} : s] \\ \prod_{i \in [m]} \ell(\Gamma_i), & \{f\} \cup_{i=1}^m \Gamma_i = \Gamma. \end{cases}$$

The chromatic cohomology of a hypergraph $\mathcal{H}$ is the cohomology of the chromatic cochain complex

$$H^{i,j}(\mathcal{H}) := \ker \delta^{i,j} / \text{Im } \delta^{i-1,j}.$$

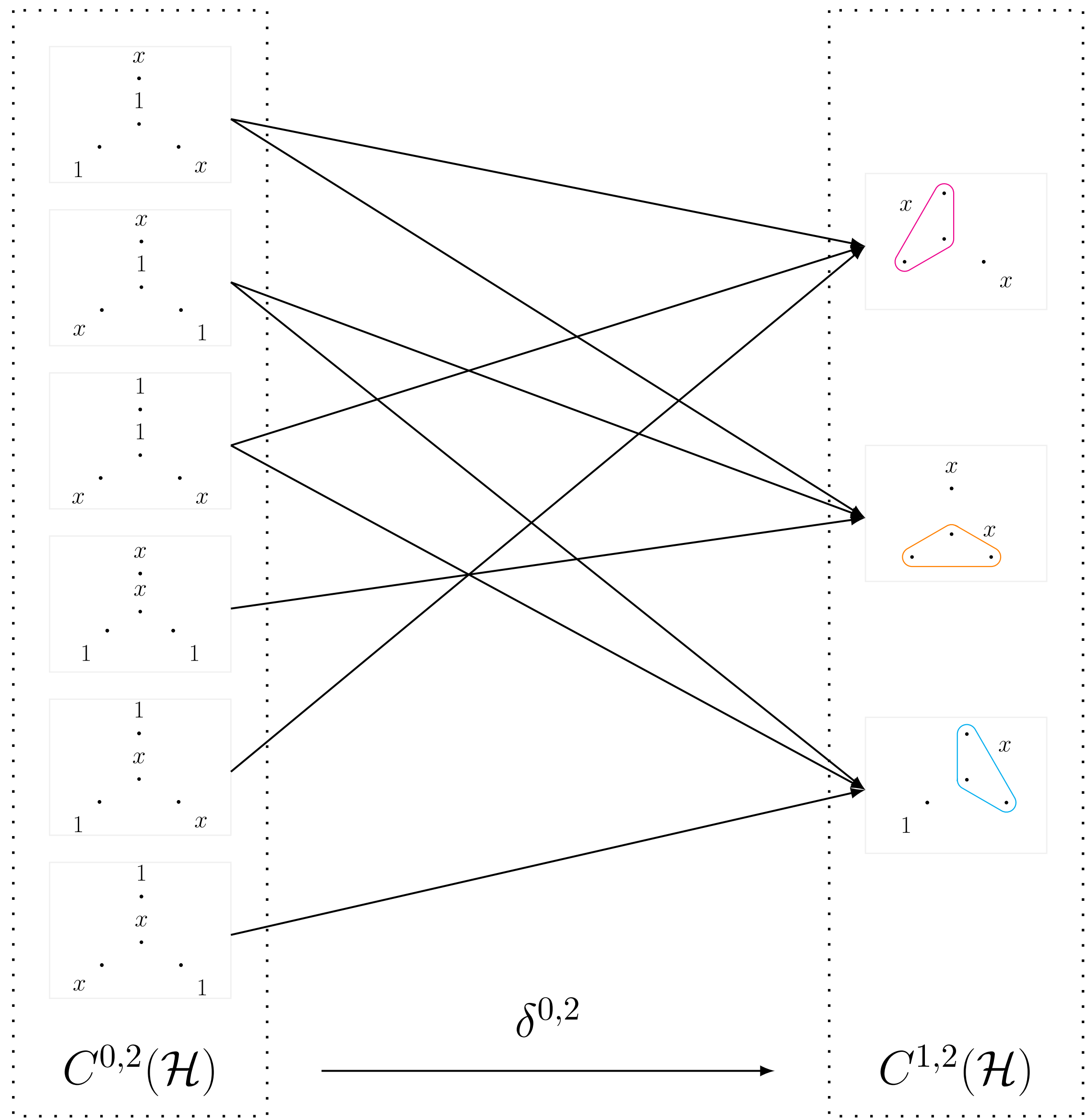


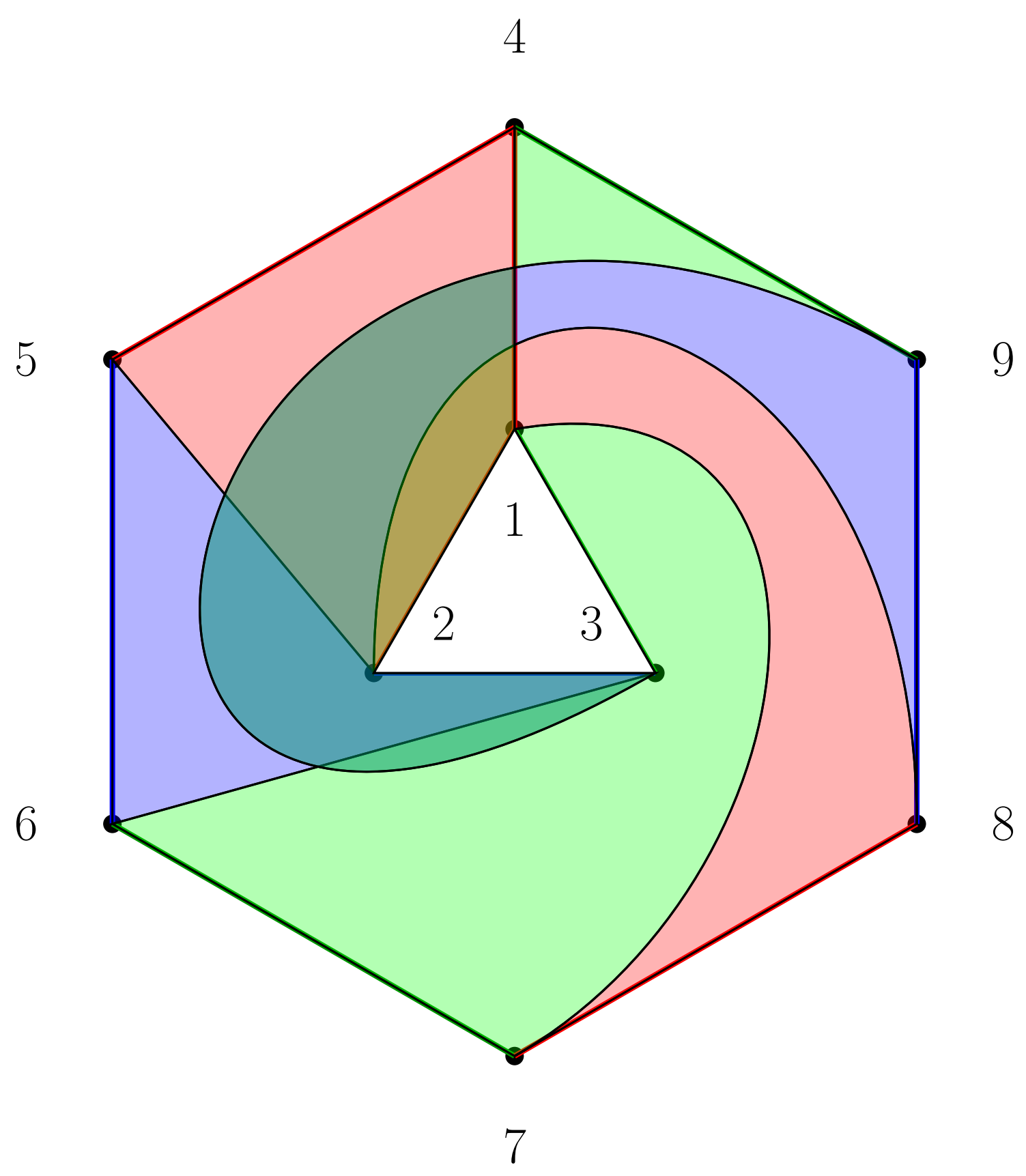
Figure 1. The coboundary map

Results

Definition The twisted pyramid $\mathcal{T}_p$ for a given odd prime p is the hypergraph on $3p$ vertices with hyperedges

$$\begin{array}{cc} \{1, 2, p+1, p+2\} & \{1, 2, 2p+1, 2p+2\} \\ \{2, 3, p+2, p+3\} & \{2, 3, 2p+2, 2p+3\} \\ \vdots & \vdots \\ \{p-1, p, 2p-1, 2p\} & \{p-1, p, 3p-1, 3p\} \\ \{1, p, 2p, 2p+1\} & \{1, p, p+1, 3p\} \\ & \{p+1, p+2, \dots, 3p\}. \end{array}$$

Theorem The chromatic cohomology group $H^{1,p}(\mathcal{T}_p)$ contains a cyclic subgroup of order p for odd primes p .



(a) $\mathcal{T}_3$

$j \backslash i$	0	1
9	$\mathbb{Z}$	$\cdot$
8	$\mathbb{Z}^9$	$\cdot$
7	$\mathbb{Z}^{36}$	$\cdot$
6	$\mathbb{Z}^{78}$	$\cdot$
5	$\mathbb{Z}^{90}$	$\cdot$
4	$\mathbb{Z}^{44}$	$\mathbb{Z}_2^6$
3	$\mathbb{Z}^2$	$\mathbb{Z}^6 \oplus \mathbb{Z}_2^8 \oplus \mathbb{Z}_3$
2	$\cdot$	$\mathbb{Z}^9$
1	$\cdot$	$\mathbb{Z}$

(b) $H^{i,j}(\mathcal{T}_3)$

Figure 2. The chromatic cohomology of $\mathcal{T}_3$

Definition Let $\mathcal{H}$ be a nonempty hypergraph. For any hyperedge $e \in \mathcal{H}$ the hypergraph $\mathcal{H}_e^n$ is the hypergraph $\mathcal{H}$ with $n \in \mathbb{Z}^+$ distinct vertices inserted into the hyperedge e . Call this new hyperedge e' .

Theorem There exists a split short exact sequence of chromatic cochain groups

$$0 \rightarrow C^{i,j}(\mathcal{H}_e^n) \xrightarrow{\mu_e \oplus \beta'} C^{i,j}(\mathcal{H}) \oplus C^{i,j}(\mathcal{H}_e^n - e') \xrightarrow{\iota_*} C^{i,j}(\mathcal{H} - e) \rightarrow 0$$

and induces a split short exact sequence on the level of cohomology

$$0 \rightarrow H^{i,j}(\mathcal{H}_e^n) \xrightarrow{(\mu_e)_* \oplus \beta'_*} H^{i,j}(\mathcal{H}) \oplus H^{i,j}(\mathcal{H}_e^n - e') \xrightarrow{\iota_*} H^{i,j}(\mathcal{H} - e) \rightarrow 0.$$

Corollary Given a hypergraph $\mathcal{H}$ the cohomology of $\mathcal{H}_e^n$ is completely determined by the cohomology of $\mathcal{H}$ and $\mathcal{H} - e$ given by the following isomorphism

$$H^{i,j}(\mathcal{H}_e^n) \oplus H^{i,j}(\mathcal{H} - e) \cong H^{i,j}(\mathcal{H}) \oplus H^{i,j}(\mathcal{H}_e^n - e) \cong H^{i,j}(\mathcal{H}) \oplus \bigoplus_{p+q=j} H^{i,p}(\mathcal{H} - e) \otimes \mathbb{Z}^{\binom{n}{q}}.$$

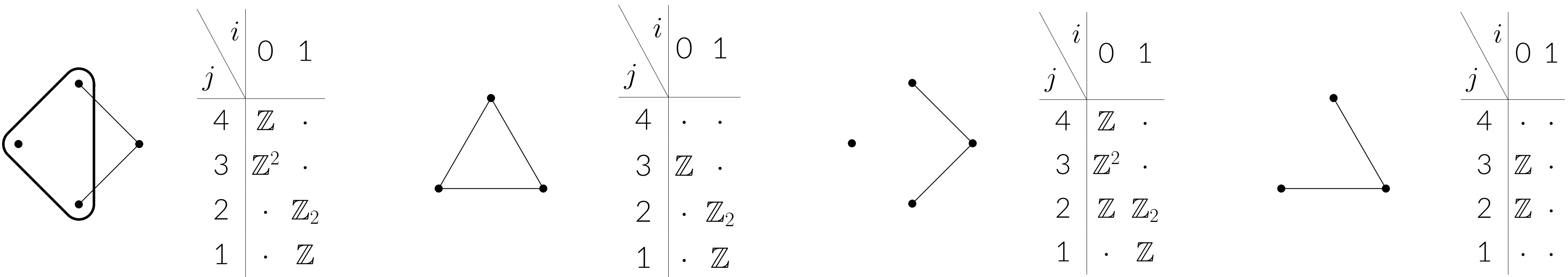


Figure 3. Inserting a vertex into the 3-cycle

Decategorification

The graded Euler characteristic of the chromatic cohomology groups of a hypergraph recovers the chromatic polynomial of the hypergraph:

$$\chi(H^{*,*}(\mathcal{H})) := \sum_{i \geq 0} \sum_{j \geq 0} (-1)^i \text{rank}(H^{i,j}(\mathcal{H})) q^j = P(\mathcal{H}; \lambda - 1).$$

The chromatic polynomial is uniquely defined by the deletion-contraction property

$$P(\mathcal{H}; \lambda) = P(\mathcal{H} - e; \lambda) - P(\mathcal{H}/e; \lambda)$$

what was categorified into the deletion-contraction long exact sequence in cohomology

$$0 \rightarrow H^{0,j}(\mathcal{H}) \rightarrow H^{0,j}(\mathcal{H} - e) \xrightarrow{\delta} H^{0,j}(\mathcal{H}/e) \rightarrow H^{1,j}(\mathcal{H}) \rightarrow \dots$$

The insertion-deletion short exact sequence decategorifies into

$$P(\mathcal{H}_e^n; \lambda) = P(\mathcal{H}; \lambda) + P(\mathcal{H}_e^n - e'; \lambda) - P(\mathcal{H} - e; \lambda)$$

which is a corollary of the deletion-contraction sequence.

Abundance of Torsion

Definition For $m, n \in \mathbb{Z}^+$ the m -uniform complete graph on n vertices is denoted $\mathcal{K}_n^m$ with hyperedge set $\binom{[n]}{m}$.

Table 1. $H^{i,j}(\mathcal{K}_5^3)$

$j \backslash i$	0	1
6	$\mathbb{Z}$	$\cdot$
5	$\mathbb{Z}^4$	$\cdot$
4	$\cdot$	$\mathbb{Z}_2^4 \oplus \mathbb{Z}_3$
3	$\cdot$	$\mathbb{Z}^5$
2	$\cdot$	$\mathbb{Z}$
1	$\cdot$	$\mathbb{Z}$

Table 2. $H^{i,j}(\mathcal{K}_6^3)$

$j \backslash i$	0	1	2
6	$\mathbb{Z}$	$\cdot$	$\cdot$
5	$\mathbb{Z}^6$	$\cdot$	$\cdot$
4	$\cdot$	$\mathbb{Z}^5 \oplus \mathbb{Z}_2^5 \oplus \mathbb{Z}_3$	$\cdot$
3	$\cdot$	$\mathbb{Z}^{15}$	$\cdot$
2	$\cdot$	$\mathbb{Z}$	$\mathbb{Z}^5 \oplus \mathbb{Z}_2^4$
1	$\cdot$	$\cdot$	$\mathbb{Z}^9$

Table 3. $H^{i,j}(\mathcal{K}_6^4)$

$j \backslash i$	0	1
6	$\mathbb{Z}$	$\cdot$
5	$\mathbb{Z}^6$	$\cdot$
4	$\mathbb{Z}^{15}$	$\cdot$
3	$\mathbb{Z}^5$	$\mathbb{Z}_2^5 \oplus \mathbb{Z}_3$
2	$\cdot$	$\mathbb{Z}^6$
1	$\cdot$	$\mathbb{Z}$

Table 4. $H^{i,j}(\mathcal{K}_7^4)$

$j \backslash i$	0	1
7	$\mathbb{Z}$	$\cdot$
6	$\mathbb{Z}^7$	$\cdot$
5	$\mathbb{Z}^{21}$	$\cdot$
4	$\cdot$	$\mathbb{Z}_2^{14} \oplus \mathbb{Z}_4 \oplus \mathbb{Z}_3^6$
3	$\cdot$	$\mathbb{Z}^{21}$
2	$\cdot$	$\mathbb{Z}^7$
1	$\cdot$	$\mathbb{Z}$

Figure 4. The chromatic cohomology of some uniform complete hypergraphs

The Goal

To discover which properties of hypergraphs are encoded by the torsion subgroups.